RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 18th June 2026	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Facilities Planning And Design		**Course Code : IM365TDA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	List any two characteristics of fixed automation systems.	(02)	1	1
2.	What is Flexible Manufacturing System (FMS)?	(02)	2	2
3.	State the concept of Just-In-Time (JIT) manufacturing.	(02)	3	3
4.	Why is a Flexible Manufacturing System (FMS) more suitable than fixed automation for product variety?	(02)	2	4
5.	Why is fixed automation not suitable for low-volume customized products? Give two reasons	(02)	2	3
	Part – B			
1.	Discuss characteristics and applications in manufacturing industries of fixed automation systems.	(10)	2	2
2.	Describe the working of a Flexible Manufacturing System (FMS).	(10)	3	2
3.	Explain the operation of a single-stage multi-machine system with a neat diagram.	(10)	3	3
4.	Illustrate how Just-In-Time (JIT) manufacturing contributes to waste elimination and productivity improvement. Explain its key principles.	(10)	3	4
5.	Propose a comprehensive plan to transform a traditional manufacturing facility into a lean production system based on JIT concepts.	(10)	2	4

****

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 17-06-2026	**CIE – III**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Service Operations Management		**Course Code : IM365TDB**

Note:
Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is 'mass customisation' and which two principles underpin it in services?	2	1	2
2.	What is the 'market-operations gap' and what risk does it pose for service organisations?	2	2	3
3.	When a bank introduces a mobile app for all routine transfers, moving the labor from the teller to the user, what role is the customer now playing?	2	2	2
4.	What is meant by customer participation in services?	2	2	4
5.	A fast-food chain uses average service time per customer as its only performance measure. What is the risk?	2	2	3
	Part – B			
1	Discuss the Hackman and Oldham Job Characteristics Model in the context of motivating service providers. How do specific job dimensions lead to better psychological states?	10	2	4
2.	Explain the different types of service processes and their importance in service operations management.	10	2	4
3.	Prestige Home Services (PHS) is a premium home maintenance company offering plumbing, electrical, carpentry, and deep-cleaning services in Hyderabad. Technicians are dispatched to customer homes and are expected to assess the problem, carry out repairs, and collect payment - all independently. PHS recently won a large corporate contract requiring 200 service visits per month across office premises. To meet this demand, PHS hired 40 new technicians rapidly with minimal on boarding. Within 3 months, customer complaints surged. Investigations revealed that technicians were arriving without knowing the scope of the job, improvising repairs without proper tools, quoting inconsistent prices to customers, and in several cases behaving rudely when customers questioned their work. Experienced technicians reported feeling demoralised as their reputation was being damaged by new hires. The operations manager is tasked with redesigning both the service process and the people management system before the corporate contract renewal in 4 months. (a) Using service blueprinting, redesign the end-to-end service process for PHS, identifying customer actions, front-stage actions, back-stage actions, support processes, and key fail points with solutions.	10	3	3

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 18th JUNE 2026	**CIE Improvement**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Design of Experiments		**Course Code : IM365TDD**

Instructions to Students:
1. Answer all questions
2. Use of Statistical tables and Orthogonal Arrays tables permitted

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is the primary purpose of a "Linear Graph" in Taguchi's orthogonal array designs?	2	2	CO3
2.	What is the "Strategy for Constructing an Orthogonal Array" when the number of factors is high but the budget is low.	2	3	CO3
3.	Illustrate an application of the Dummy Level Technique for a factor with 2 levels in an L_9 (3-level) array.	2	2	CO3
4.	What is the primary purpose of a "Verification Experiment" in the robust design cycle?	2	2	CO2
5.	Contrast "Testing Conditions" with "Control Factor Levels" in a typical study.	2	2	CO2
	Part – B			
1	In an investigation on the stability of power factor of capacitor it was decided to examine the following factors: **Sl No** / **Factor** / **Levels** 1 / Leakage Current of Foil (A) / $\leq$ 2 mA / $\geq$ 2 mA 2 / Power Factor of Foil (B) / $\leq$ 3% / $\geq$ 3% 3 / Capacitance Value of Coil © / 8-10 μf / 11-13 μf 4 / Soaking Time during Impregnation / 2 hrs / 4 hrs 5 / Loading Pattern / 70 % / 100 % Suspected interaction effects are AB, BC. Develop the Orthogonal Array Layout and experimental plan.	15	4	CO3

<table>
<tr><td>2</td><td>In a study to examine process control parameter for joining “BUS-BAR” of an electrical system the following factors are levels were identified for investigation with respect to efficiency of joining
<table>
<tr><th rowspan="2">Factor Code</th><th rowspan="2">Factor Description</th><th colspan="2">Level</th></tr>
<tr><th>1</th><th>2</th></tr>
<tr><td>A</td><td>Torque</td><td>25 ft lb</td><td>40 ft lb</td></tr>
<tr><td>B</td><td>Compound</td><td>Nil</td><td>Silicon Grease</td></tr>
<tr><td>C</td><td>Surface Finish</td><td>As is</td><td>Ground</td></tr>
<tr><td>D</td><td>Joint Overlap</td><td>8 mm</td><td>16 mm</td></tr>
</table>
Develop an orthogonal array design to estimate all the main effects and interaction effects A X B and C X D.</td><td>15</td><td>4</td><td>CO3</td></tr>
<tr><td>3</td><td>Provide a concise summary of major Shainin Techniques.</td><td>10</td><td>2</td><td>CO2</td></tr>
<tr><td>4</td><td>Enumerate the four types of Signal-to-Ratios for the static problems</td><td>10</td><td>2</td><td>CO2</td></tr>
</table>

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 25th June 2025	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
IM365TDA - FACILITIES PLANNING and DESIGN		

Note:
1. Answer all the Questions.

PART A

Q No.1	Questions	M	BT	CO
1.1	State one trend in facilities planning aligned with Industry 4.0.	02	2	CO1
1.2	What does a pull system in JIT manufacturing imply?	02	1	CO1
1.3	Why is reducing Work-In-Process (WIP) important in manufacturing?	02	2	CO1
1.4	List the storage operations.	02	1	CO4
1.5	Write two principles of order picking.	02	2	CO4

Part –B

Sl. No	Questions	M	BT	CO
1	Explain the concept of Flexible Manufacturing Systems (FMS). How do they improve production efficiency compared to fixed automation?	10	L2	CO1
2	Differentiate between single-stage multi-machine systems and multi-stage systems. Give examples of each.	10	L3	CO3
3	Define fixed automation systems. What are their advantages and limitations in mass production environments?	10	L2	CO2
4	Sketch and explain the functions in the warehouse.	10	L3	CO4
5	Enumerate and elaborate shipping principles.	10	L1	CO4